THIAGO GONÇALVES

PAINTER + CREATIVE DEVELOPER

Curitiba, Brazil · remote, any time zone zednaked@gmail.com zednaked.github.io/portfolio github.com/zednaked
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PROFILE

Creative developer who ships interactive work to the browser and measures it: a three.js flood simulation running 2.4 million water cells on a laptop GPU, a synth firmware compiled to WebAssembly, games served at $0 a month, and, as sole front-end engineer, a catalogue of 20+ commercial titles exported to the web under strict bandwidth budgets. I care about the frame, the byte and the person on the other end of a slow phone. Before code I was a painter, and I still am: I bring motion, timing and composition from fourteen years of audiovisual production. Professional React since 2014: I built and ran the platform agencies and advertisers used to deliver TV commercials to air. I use Claude Code every day and build my own tools around it.

EXPERIENCE

Senior Game Developer, sole front-end engineer Sep 2025 – present
Freelance contract · confidential client under NDA · remote

- Sole front-end engineer for a catalogue of **20+ commercial Godot titles shipped to the browser**.
- Shared runtime architecture replicated across 17 projects from a common template: event bus, API layer, Socket.IO client, i18n, audio, telemetry; semantic versioning per release.
- Web export under hard bandwidth budgets: catalogue-wide WebP conversion, lossy import taking one build **from 119 MB to 37 MB**, audio streamed outside the pack, service workers with a purpose-built test harness. Sixteen production builds cut 23% to 76% with no assets deleted.
- Internationalization to **12 locales including Arabic RTL**, font fallback for Hindi and Nepali, localized currency and voice-over.
- Optimistic UI with server reconciliation, session recovery and tab-switch resynchronization.
- Contributed to the **Vue 3 shell** (Vite, vue-i18n, Socket.IO) that hosts the catalogue in the browser.
- Turned a designer's mockups into a **playable game in a single day**.
- 9 custom GLSL shaders and Spine 2D runtime integration across the catalogue.
- **1,082 commits · 172 issues delivered · 110 pull requests · ~103k lines of GDScript** in twelve months.

Game Developer, freelance projects 2019 – 2024
Remote, with international teams

- **UrbanSim** (2024): interactive urban planning simulator in Godot, 3D city with simulated traffic and weather; presented at an urban planning congress.
- **Evorse** (2023): game MVP in Godot, from 2D art to mechanics, menus and HUD.
- **EduXR** (2022) and **Realms of Reality** (2020): hybrid VR and AR narratives for Oculus Quest with haptic feedback.
- **Pixel Quest** (2019): 2D platformer, 20 levels of hand-drawn art; beta with 500+ players tuned through analytics.

CEO and full-stack developer 2007 – 2021
Control-C · audiovisual and media production · Curitiba

- **TV commercial delivery platform (2017–2020), React front end and Python back end, whole stack by me**: agencies, production houses and advertisers uploaded the spots that went to air, and it handled a large share of the material delivered to the Globo network affiliate in southern Brazil.
- **Video segment intake system (2014–2015), React**: local production houses uploaded footage for the company to edit.
- Fourteen years running a production company: client briefs, art direction, editing and motion, delivery.

SELECTED WORK

Portfolio site

React · TypeScript · GSAP · WebGL · three.js · Vite

A wet-paint WebGL opening, a three.js terrain whose water rises with the scroll and the hand, chapter cards that collapse into their shape; gentler under reduced motion. **Lighthouse 98 on a phone (LCP 2.0 s), 100 on desktop, CLS 0**; WCAG 2.2 AA, 36 automated tests and a Lighthouse gate on every deploy.

Cheia, flood simulator

three.js · GLSL · Vite · Service Worker

Rain runs over the real terrain of a metropolitan region in a browser tab. **2.4 M cells, one shader pass per step, 5.9 ms a step, 81 fps** on an integrated GPU.

Ciberteia, multiplayer RPG

Godot web · shaders · TypeScript · Cloudflare Workers, Durable Objects, D1

Zero art files; hibernating WebSockets took requests from **720 to 1 per player-hour**, at $0 a month.

Nirbija, a Linux mixer that is also an instrument

C++20 · Qt Quick · JACK / PipeWire · LV2, CLAP, VST3

A rack of channel strips with a drone synth, sampler, looper and step sequencer built in, plugins hosted with their own editors. **16 strips rendered in 0.1 ms, 2% of the block**; no allocation, lock or log on the audio thread.

Jangada Studio

WebAssembly · AudioWorklet · Web MIDI · Canvas

The firmware of a pocket synth compiled to WebAssembly and played in the browser: **13 synth engines**, the same code as the hardware.

Playable ad under the cap

PixiJS v8 · WebGL · esbuild

One mechanic in a single HTML file: **630 KB, 30.8% of Meta's 2 MB cap**.

SHIPPED AND CREDITED

- **Felucca** (synth firmware): my ratchets shipped in release 1.0.5, credited in the notes. hugelton/Felucca #100
- **Sonix** (Rust DAW): found an unrouted plugin output growing ~384 KB a second by reading the code; fixed and credited the same day. sonix cdb2447
- **Build budget write-up**: the method behind the 23% to 76% cuts. godot-web-build-budget

SKILLS

Front-end	JavaScript, TypeScript, React (since 2014), Vue 3, HTML, CSS, Vite, esbuild; responsive layout, accessibility (reduced motion, keyboard, focus, alt text), Core Web Vitals
Motion and graphics	GSAP (ScrollTrigger, Flip), CSS animation, three.js, WebGL, GLSL shaders, Canvas, PixiJS, Spine 2D
Engines and runtime	Godot 4 (GDScript, web export), C++20 real-time audio, Qt Quick / QML, WebAssembly, AudioWorklet, Web MIDI, service workers
Back end and infra	Python, Cloudflare Workers, Durable Objects, D1, R2, WebSocket, Socket.IO, GitHub Actions, Linux
Performance	bundle and asset budgets, texture and audio compression, frame-time profiling, LCP and CLS measurement
AI-assisted work	Claude Code daily, with my own tools and agents around it

ART

- **2015**, Carrousel du Louvre, Paris: gold medal, Artcom Expo International (oil portrait).
- **2015**, Embassy of Brazil, Oslo: medal and honour of merit; the painting stayed with the embassy.
- **2015**, Artcom Expo, Dubai: honourable mention; in the book *Artists Across Continents 2015*.
- **2014**, solo show and portrait lecture, Pinhais · **2021**, *Kleshas*, Fundação Cultural de Curitiba.

LANGUAGES

Portuguese, native · English, fluent · Spanish, advanced · German, Italian, basic

WORK SETUP

Remote from Curitiba (UTC−3), overlapping US hours. US clients: mentoring and consulting, under NDA.

EDUCATION

B.Des. Product Design, Univ. Tuiuti do Paraná, 2009